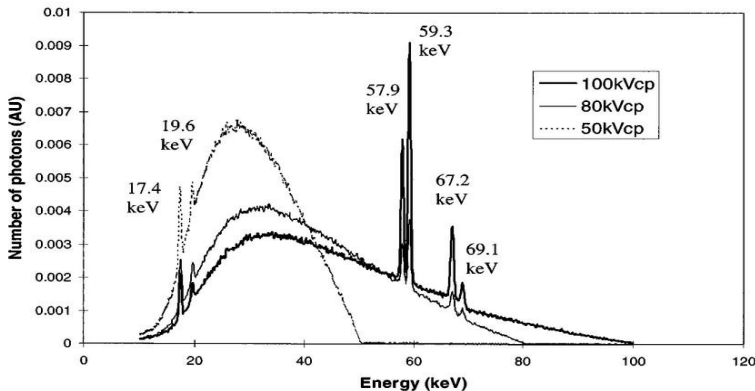


7	
(a)(i)1.	$E = hf = h \frac{c}{\lambda}$ $= \frac{6.63 \times 10^{-34} (3.00 \times 10^8)}{250 \times 10^{-9}}$ $= 7.96 \times 10^{-19} \text{ J}$ $= 4.97 \text{ eV}$
	$hf = hf_0 + eV_s$ $4.97 = hf_0 + 1.20$ $hf_0 = 3.77 \text{ eV}$
	$= 3.8 \text{ eV}$
(a)(i)2.	If <u>frequency/energy/wavelength of photon remains constant</u> ,
	when <u>intensity changes</u> , implies that ONLY the <u>number of photons per unit time</u> arriving at the emitter <u>changes</u> .
	Using $hf = hf_0 + eV_s$ • As the electrode is the same, work function ϕ is constant. • Each photon imparts the same energy hf to each electron • eV_s remains the same, hence V_s remains the same.
(a)(ii)1.	Only one electrode is emitting photoelectrons.
	In (a)(i), there is current only the positive direction, implies only one electrode is emitting electrons. As work function of zinc (electrode X) is lower than 250 nm, electrons must be <u>emitted from X only</u> .
(a)(ii)2.	As the current flows in 2 directions, both <u>electrodes are emitting electrons</u> , which implies that photon energy is greater than the work functions of X and Z.
	<u>Negative values</u> of photocurrent is caused by photoemission <u>from Z</u> .
(b)(i)1.	Same characteristic peaks at higher wavelength.
	Same target (same element).
(b)(i)2.	More photons are produced over the continuous spectrum and peaks (higher intensity). OR A has a lower minimum wavelength.
	A has higher accelerating potential (electrons of A has higher kinetic energies).
(b)(ii)	From the graph, $\lambda_{\min} = 16 \times 10^{-12} \text{ m}$
	$\frac{1}{2}mv^2 - 0 = qV \dots\dots(1)$ $\frac{1}{2}mv^2 - 0 = hf_{\max} \dots\dots(2)$ $qV = h \frac{c}{\lambda_{\min}}$ $(1.60 \times 10^{-19})V = \frac{6.63 \times 10^{-34} (3.00 \times 10^8)}{16 \times 10^{-12}}$
	$V = 77700 \text{ V}$

(b)(iii)	
	Energy difference of levels Tungsten : A (57.4 keV), B (59.3 keV) Molybdenum: P (17.4 keV), Q (17.5 keV)
(b)(iv)	In Fig. 7.4, The taller peaks (57.9, 59.3) is <u>similar to the energy difference of levels</u> of tungsten. (There is a small difference between calculations and the peaks in the graph due to uncertainties in measurements.) ➤ Target contains tungsten.
	The shorter peaks (17.4) <u>matches with the energy difference of levels</u> of molybdenum. (But the peak (17.5) is not shown. It is because the difference between 17.4 and 17.5 is very small and hence both may be <u>represented by just one peak.</u>) ➤ Target contains molybdenum.
	Since peaks due to tungsten is much taller than that of molybdenum, ➤ the target is made of <u>mainly</u> tungsten and <u>less</u> molybdenum.